JBREC
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Heart Stroke Prediction System



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Abstract

This abstract outlines a Heart Stroke Prediction system that leverages machine learning to accurately assess the likelihood of an individual developing heart disease based on their medical history. The system utilizes algorithms like logistic regression and KNN, demonstrating its ability to effectively classify patients and improve prediction accuracy compared to traditional methods.

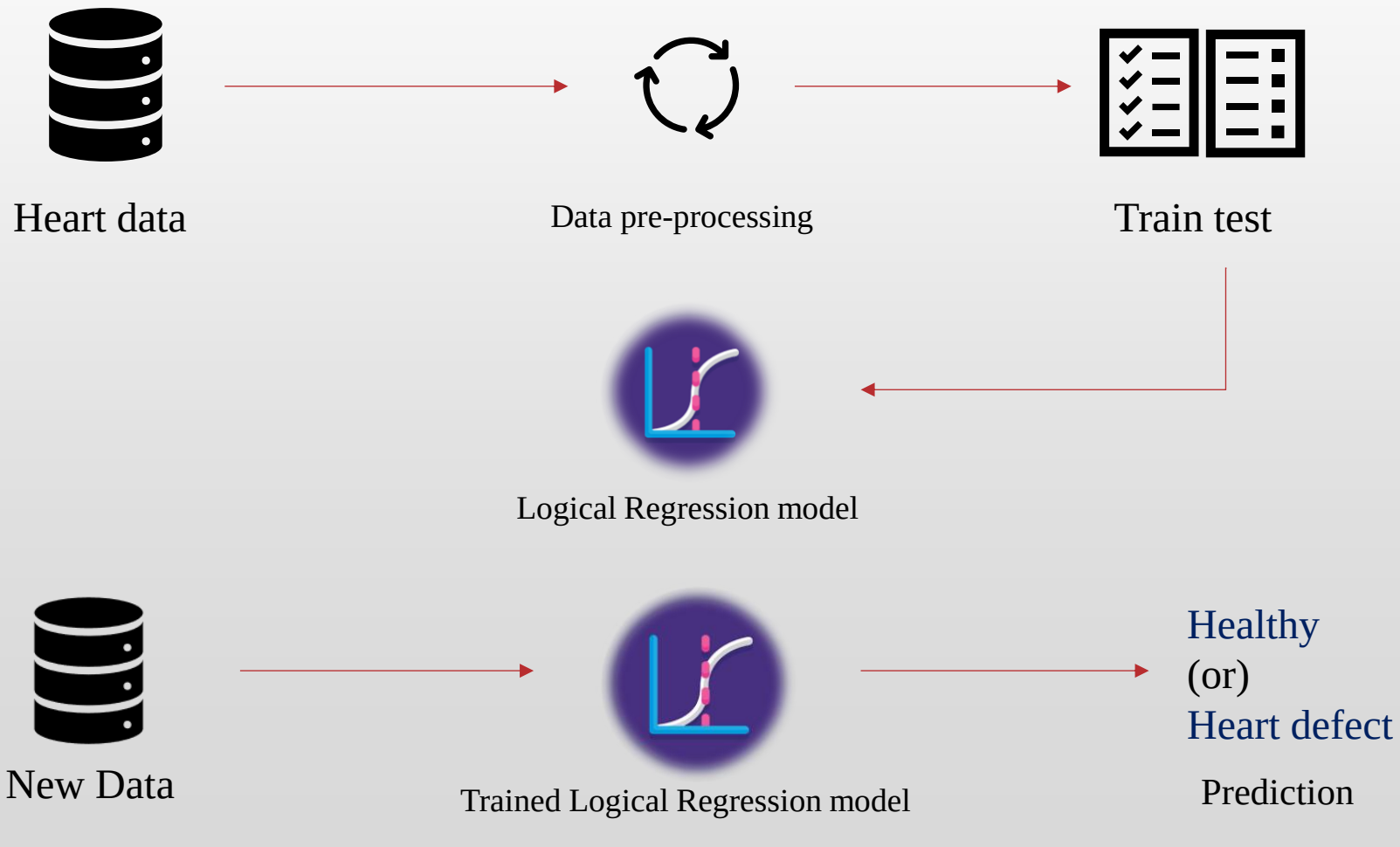
Key points about the heart disease prediction system:

- **Focus:** Predicting the risk of heart disease in individuals based on medical data.
- **Methods:**
  - Employs machine learning algorithms like logistic regression and KNN.
  - Demonstrates improved accuracy compared to other classifiers like Naive Bayes
  - Utilizes patient medical history to make predictions.
- **Goal:**
  - To provide a reliable and efficient system for diagnosing and potentially preventing heart disease

Field of study

- **Domain:**
  - The domain for Heart Stroke Prediction encompasses several areas, including demographic data, clinical information, laboratory results, and medical history. These factors are used to build predictive models that identify individuals at risk of developing heart disease.

Deliverables



Beneficiaries of the project

- **Patients:**
  - Early risk identification, allowing for preventive measures
- **Doctors & clinicians:**
  - A decision support tool to aid in diagnosing and assessing stroke risk.
- **Hospitals & Healthcare Providers:**
  - Improved resource allocation.



Progress of the study

- **Dataset Acquisition & Understanding.**
  - Source of Dataset : Kaggle
  - Key features:age,gender,bmi,smoking_status,avg_BP.
- **Data Preprocessing:**
  - Handling Misssing values.
  - Feature Scaling Normalization
- **Model Implementation:**
  - Logistic Regression.
  - K-Nearest Neighbors (KNN).
  - Decision Tree
- **Traing and Evaluation.**

Modules

- **Data Input & Preprocessing Module**:
  - Takes raw patient data.
  - Performs data Pre-processing.
- **Model Training Module:**
  - Contains implementation of Logistis Regression, Decision tree.
  - Train the preprocessed dataset.
  - Saves the trained models.
- **Prediction Module:**
  - Loads a selected trained model.
  - Takes new data as input.
- **Evaluation & Reporting:**
  - Uses test data to valuate and outputs stokes(0/1)

Thank You